

Assignment 3&4

Due on April 30, 2026

平行板电容器

1. (a) Consider the parallel plate capacitor shown in Figure Q1a where the left plate is at 0V, and the right plate is biased at +2.0 V. An electron is originally at rest in vacuum just beyond the surface of the left-hand plate. Starting from the equation of motion $F = m \cdot a$ or otherwise, determine the time taken for the electron to reach the right-hand plate, and its velocity by the time it reaches the right-hand plate. The free-electron mass is $m_0 = 9.108 \times 10^{-31}$ kg. The force F on a charged-particle of charge Q in an electric field E is $F = Q \cdot E$. (20 pts)
- (b) If a uniform metal with a workfunction of 5.0 eV were placed and connected between the biased parallel plates (Figure Q1b), sketch the energy-band diagram of the metal within the plate, indicating the filled and empty parts of the band, the vacuum level, and the workfunction. (20 pts)
- (c) Consider an electron in the metal of part (b) at the top of the filled part of the band, starting at the left-hand side. In a highly-simplified model*, assume that the electron undergoes a scattering event every 10 fs ($1 \text{ fs} = 10^{-15} \text{ s}$), in which the electron loses all its acquired kinetic energy.
 - (i) From the equation of motion, determine the average velocity, v_d , of the electron assuming that it gains kinetic energy as it accelerates in the electric field between scattering events. How long will it take to reach the right-hand side? (10 pts)
 - (ii) A parameter, known as mobility μ , relates the average velocity v_d to the electric field E , via $v_d = -\mu E$. (the minus sign takes into account the fact that the negatively-charged electron moves in a direction opposite to the electric field). What is the mobility of the electron? Express your answer in units of $\text{cm}^2 / \text{V} \cdot \text{s}$ (10 pts)

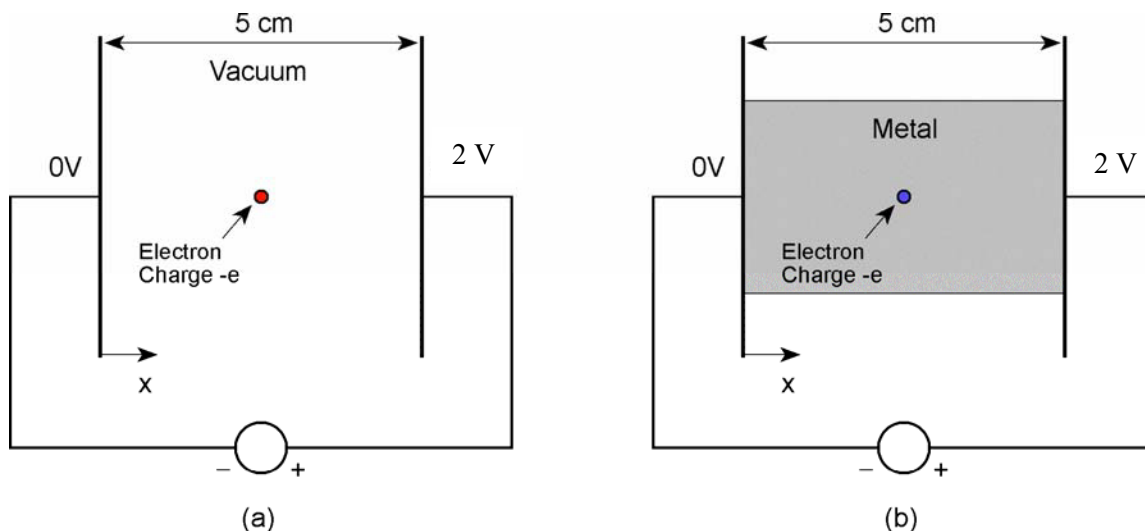


Figure Q1

2. Determine the temperature at which 80 percent of acceptor atoms are ionized. Consider p-type silicon doped with boron at a concentration of $N_a = 5 \times 10^{16} \text{ cm}^{-3}$. (20 pts)

硼

1. (a) 真空中的电子运动

$$d = 5 \text{ cm} = 0.05 \text{ m}$$

$$\Delta V = 2.0 \text{ V} - 0 \text{ V} = 2.0 \text{ V}$$

$$m_0 = 9.108 \times 10^{-31} \text{ kg} \quad e = 1.602 \times 10^{-19} \text{ C}$$

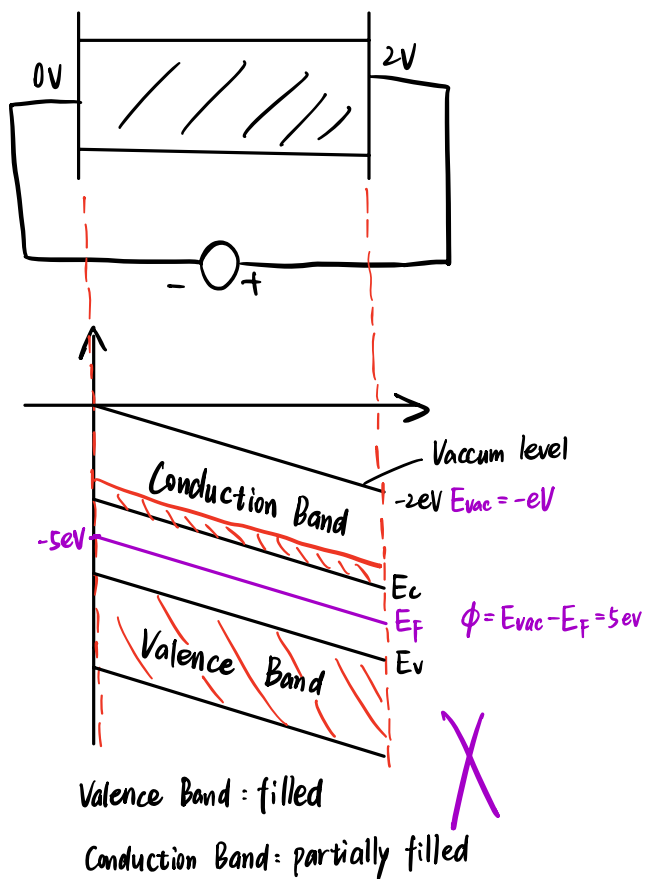
$$E = \frac{\Delta V}{d} = \frac{0.2}{0.05} = 40 \text{ V/m}$$

$$a = \frac{eE}{m} = \frac{1.602 \times 10^{-19} \times 40}{9.108 \times 10^{-31}} \approx 7.035 \times 10^{12} \text{ m/s}^2$$

$$t = \sqrt{\frac{2d}{a}} = \sqrt{\frac{2 \times 0.05}{7.035 \times 10^{12}}} \approx 1.192 \times 10^{-7} \text{ s}$$

$$v = at = (7.035 \times 10^{12}) \times (1.192 \times 10^{-7}) \approx 8.386 \times 10^5 \text{ m/s}$$

(b) 绘制金属能带图



(c) 金属中的电子输运(散射模型)

(i) 平均速度与到达时间

电子每次碰撞后动能归零

两次碰撞间做初速为0的匀加速运动

$$v_{max} = a\tau = (7.035 \times 10^{12}) \times 10^{-14} \approx 0.07035 \text{ m/s}$$

$$v_d = \frac{1}{2} a\tau = 0.035175 \text{ m/s}$$

$$T = \frac{d}{v_d} = \frac{0.05 \text{ m}}{0.035175 \text{ m/s}} \approx 1.421 \text{ s}$$

(ii) 迁移率 μ

$$\mu = \frac{v_d}{E} = \frac{0.035175 \text{ m/s}}{40 \text{ V/m}} = 8.79375 \times 10^{-4} \text{ m}^2/(\text{V}\cdot\text{s})$$

$$= 8.79 \text{ cm}^2/(\text{V}\cdot\text{s})$$

2. 求受主电离比例达80%时的温度

已知 1. $N_A = 5 \times 10^{16} \text{ cm}^{-3}$

2. $p = N_A^- = 80\% N_A = 4 \times 10^{16} \text{ cm}^{-3}$

3. 对于硅中的硼, 受主电离能约为 $E_a - E_v = 0.045 \text{ eV}$

4. 受主状态的简并因子取 $g_a = 4$ (硅的价带特征)

5. 价态有效态密度 $N_v = 1.04 \times 10^{19} \times \left(\frac{T}{300}\right)^{1.5} \text{ cm}^{-3}$

degeneracy factor

电离统计 $N_A^- = \frac{N_A}{1 + g_a \exp\left(\frac{E_a - E_F}{kT}\right)}$

$$\Rightarrow 0.8 = \frac{1}{1 + 4 \exp\left(\frac{E_a - E_F}{kT}\right)} \Rightarrow E_F - E_a = 2.77 kT$$

空穴浓度 $p = N_v \exp\left(-\frac{E_F - E_v}{kT}\right)$

$$\Rightarrow 4 \times 10^{16} = N_v(T) \cdot e^{-2.77} \cdot \exp\left(-\frac{E_a - E_v}{kT}\right) \Rightarrow 0.015 = \left(\frac{T}{300}\right)^{1.5} \exp\left(-\frac{522}{T}\right)$$

迭代法求T

当 $T = 220 \text{ K}$ 时, 右边 ≈ 0.058

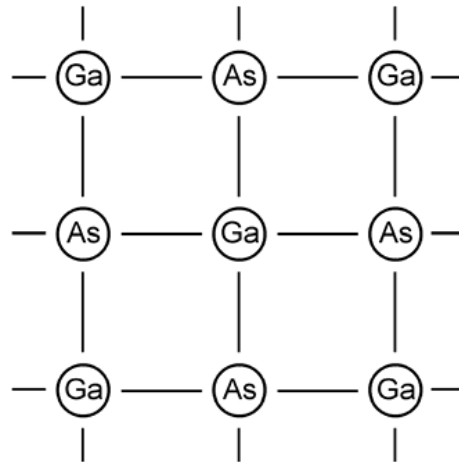
当 $T = 230 \text{ K}$ 时, 右边 ≈ 0.071

$\Rightarrow$ 插值得 $T \approx 223 \text{ K}$

Elemental $\hookrightarrow$ 单质半导体

化合物半导体

3. Gallium arsenide (GaAs) is a compound semiconductor made out of equal numbers of gallium (Ga – group III) and arsenic (As – group V) atoms, as shown schematically. Its intrinsic carrier concentration at 300 K is 10^7 cm^{-3} while its bandgap is 1.42 eV.



Silicon (group IV) is used to dope GaAs. Silicon can either replace Ga atoms or As atoms in GaAs. On average, 90% of the silicon atoms replaces Ga atoms, while the remaining 10% replaces As atoms. A GaAs sample is uniformly doped with 10^{12} cm^{-3} Si atoms.

- Is the doped GaAs n-type or p-type? Explain. (10 pts)
- Determine the equilibrium electron and hole concentrations at 300 K. State and verify any assumptions made. (10 pts)
- Determine the position of the Fermi level relative to the intrinsic Fermi level. (10 pts)

4. A silicon sample at 300 K contains 10^{16} cm^{-3} donors and 10^{18} cm^{-3} acceptors. Use $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$

- Assume that all donors and acceptors are ionized. Find the equilibrium electron and hole concentrations and the position of the Fermi level relative to the valence band edge. (10 pts)
- If a donor atom is ionized, is the corresponding donor state occupied or unoccupied? If an acceptor atom is ionized, is the corresponding acceptor state occupied or unoccupied? (10 pts)
- The donor level is 0.04 eV below the conduction band edge. From the location of the Fermi level determined in part (a) above, determine the concentration of donors that are not ionized. Assume that each donor atom represents one state at the donor level. (10 pts)
- Explain how the Fermi level changes as the sample is heated up to 1000K. (10 pts)

END

As T increases, the intrinsic carrier concentration $n_i(T)$ grows exponentially

When $n_i \gg 10^{18} \text{ cm}^{-3}$ the semiconductor enters the intrinsic region

conduction band electrons and valence band holes originate almost entirely from intrinsic thermal excitation

To satisfy the charge neutrality condition $n \approx p \approx n_i$

E_F will shift upward from near E_v to near the middle of the bandgap

3. (a) 判断掺杂类型

$$Ga: N_d = 0.9 \times 10^{12} \text{ cm}^{-3} \quad As: N_a = 0.1 \times 10^{12} \text{ cm}^{-3}$$

$N_d > N_a$ 为 n 型

(b) 平衡载流子浓度

假设: 在 300K 下杂质完全电离

且由于本征浓度 $n_i = 10^7 \text{ cm}^{-3}$ 极小, 忽略本征激发的影响

$$\text{电子浓度: } n_0 \approx N_d - N_a = 9 \times 10^{11} - 1 \times 10^{11} = 8 \times 10^{11} \text{ cm}^{-3}$$

$$\text{空穴浓度: } p_0 = \frac{n_i^2}{n_0} = \frac{(10^7)^2}{8 \times 10^{11}} = 125 \text{ cm}^{-3}$$

验证: $n_0 \gg n_i$ ($8 \times 10^{11} \gg 10^7$), 假设成立 Validation

(c) 费米能级位置

$$\text{由 } n_0 = n_i \exp\left(\frac{E_F - E_i}{kT}\right)$$

在 300K 时, $kT \approx 0.0259 \text{ eV}$

$$E_F - E_i = kT \ln\left(\frac{n_0}{n_i}\right) = 0.0259 \times \ln\left(\frac{8 \times 10^{11}}{10^7}\right) \approx 0.292 \text{ eV}$$

$\Rightarrow$ Fermi level 位于 intrinsic Fermi level 上方 0.292 eV 处

4. (a) 平衡载流子浓度 + 费米能级位置

$$N_d = 10^{16} \text{ cm}^{-3} \ll N_a = 10^{18} \text{ cm}^{-3} \quad \text{为 p 型半导体}$$

$$\text{净受主浓度: } N_a^- = N_a - N_d = 9.9 \times 10^{17} \text{ cm}^{-3} \quad \text{Net concentration}$$

$$\text{空穴浓度: } p_0 \approx N_a - N_d = 9.9 \times 10^{17} \text{ cm}^{-3}$$

$$\text{电子浓度: } n_0 = \frac{(1.5 \times 10^{10})^2}{9.9 \times 10^{17}} \approx 2273 \text{ cm}^{-3}$$

$$\text{费米能级位置: } E_F - E_v = k_B T \ln\left(\frac{N_v}{p_0}\right) = 0.0259 \times \ln\left(\frac{1.04 \times 10^{19}}{9.9 \times 10^{17}}\right) \approx 0.061 \text{ eV}$$

(b) 电离状态占据情况

施主能级: $D^0 \rightarrow D^+ + e^-$ 电离后未被电子占据

受主能级: $A^0 + e^- \rightarrow A^-$ 电离后未被电子占据

(d) 加热时费米能级变化

随 $T \uparrow$ $n_i(T)$ 指数级迅速增长

当 $n_i \gg 10^{18}$ 时, 半导体进入本征区, 导带电子和价带空穴几乎全来源于本征激发

为满足 $n \approx p \approx n_i$ 的平衡条件, E_F 会从靠近 E_v 边缘处上移, 最终停在带隙中间附近 (靠近 E_i)

(c) 未电离施主浓度

$$\text{施主能级: } E_d = E_c - 0.04 \text{ eV}$$

$$\Rightarrow E_d - E_F = (E_c - E_v) - (E_F - E_v) - 0.04$$

$$= 1.12 - 0.061 - 0.04 = 1.019 \text{ eV}$$

每个原子一个状态, 用费米分布函数

$$n_{\text{un-ionized}} = N_d \cdot f(E_d) = N_d \cdot \frac{1}{1 + \exp\left(\frac{E_d - E_F}{k_B T}\right)}$$

$$= 10^{16} \cdot \frac{1}{1 + \exp\left(\frac{1.019}{0.0259}\right)} \approx 10^{16} \cdot 10^{-17.3} \approx 0 \text{ cm}^{-3}$$

由于费米能级远在下

施主能级被占据的概率极低, 几乎全电离

知识点

一、载流子浓度与统计

1. 状态密度分布 $n_0 = N_c \exp\left(-\frac{E_c - E_F}{k_B T}\right)$

$$p_0 = N_v \exp\left(-\frac{E_F - E_v}{k_B T}\right)$$

2. 本征浓度与质量作用定律

$$n_i^2 = n_0 p_0$$

$$n_i = \sqrt{N_c N_v} \exp\left(-\frac{E_0}{2k_B T}\right)$$

3. 费米能级位置

对本征能级: $E_F - E_i = k_B T \ln\left(\frac{n_0}{n_i}\right)$

对价带顶(p型): $E_F - E_v = k_B T \ln\left(\frac{N_v}{p_0}\right)$

4. 补偿半导体: 同时掺杂施主、受主, 净浓度决定极性

5. 本征化: 当T很高时, $n_i \gg N_A(N_D)$

三、载流子输运

1. 漂移: 电场 $E = \frac{V}{L}$

漂移速度 $v_d = \mu E$

电流密度 $J_{drift} = (qn\mu_n + qp\mu_p)E$

2. 扩散 $J_{n,diff} = qD_n \frac{dn}{dx}$

$$J_{p,diff} = qD_p \frac{dp}{dx}$$

3. 爱因斯坦关系: $\frac{D}{\mu} = \frac{k_B T}{q}$

4. 马西森定则: $\frac{1}{\mu} = \frac{1}{\mu_c} + \frac{1}{\mu_i}$

5. 散射 $\left\{ \begin{array}{l} \text{晶格散射: 高温主导, } \mu \propto T^{-3/2} \\ \text{杂质散射: 低温、重掺杂, } \mu \propto T^{3/2}/N_{total} \end{array} \right.$

二、杂质电离

1. $\left\{ \begin{array}{l} \text{受主电离比例: } \frac{N_a^-}{N_a} = \frac{1}{1 + g_a \exp\left(\frac{E_a - E_F}{k_B T}\right)} \\ \text{施主电离比例: } \frac{N_d^+}{N_d} = \frac{1}{1 + g_d \exp\left(\frac{E_d - E_F}{k_B T}\right)} \end{array} \right.$

2. 冻结效应 (Freeze-out): 低温下杂质不能完全电离
载流子浓度锐减

3. 电离本质: $\left\{ \begin{array}{l} \text{施主失去电子} \\ \text{受主捕获电子} \end{array} \right.$

四、能带与界面

1. 功函数 $\Phi = E_{vac} - E_F$

2. 偏置电压 $\Delta E = -q \cdot \Delta V$ (正电压让能带下移)

3. 费米能级对齐: 系统达热平衡唯一标志
真空能级倾斜: 代表空间存在静电场

4. 金属-金属接触

五、常用参数 (Si 300K)

热能	$k_B T/q$	0.0259 V
带隙	E_g	1.12 eV
本征浓度	n_i	$1.5 \times 10^{10} \text{ cm}^{-3}$
有效态密度	N_c, N_v	$\sim 10^{19} \text{ cm}^{-3}$
掺杂电离能	ΔE	0.04 - 0.05 eV